



```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: df = pd.read_csv("student_exam_score.csv")
df
```

```
Out[3]:
```

	Unnamed: 0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	Par
0	0	female	NaN	bachelor's degree	standard	none	
1	1	female	group C	some college	standard	NaN	
2	2	female	group B	master's degree	standard	none	
3	3	male	group A	associate's degree	free/reduced	none	
4	4	male	group C	some college	standard	none	
...	...	...	...	...	...	...	...
30636	816	female	group D	high school	standard	none	
30637	890	male	group E	high school	standard	none	
30638	911	female	NaN	high school	free/reduced	completed	
30639	934	female	group D	associate's degree	standard	completed	
30640	960	male	group B	some college	standard	none	

30641 rows × 15 columns

```
In [4]: df.describe()
```

Out[4]:	Unnamed: 0	NrSiblings	MathScore	ReadingScore	WritingScore
count	30641.000000	29069.000000	30641.000000	30641.000000	30641.000000
mean	499.556607	2.145894	66.558402	69.377533	68.418622
std	288.747894	1.458242	15.361616	14.758952	15.443525
min	0.000000	0.000000	0.000000	10.000000	4.000000
25%	249.000000	1.000000	56.000000	59.000000	58.000000
50%	500.000000	2.000000	67.000000	70.000000	69.000000
75%	750.000000	3.000000	78.000000	80.000000	79.000000
max	999.000000	7.000000	100.000000	100.000000	100.000000

In [6]: `print(df.head())`

	Unnamed: 0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	\
0	0	female	NaN	bachelor's degree	standard	none	
1	1	female	group C	some college	standard	NaN	
2	2	female	group B	master's degree	standard	none	
3	3	male	group A	associate's degree	free/reduced	none	
4	4	male	group C	some college	standard	none	

	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	\
0	married	regularly	yes	3.0	school_bus	
1	married	sometimes	yes	0.0	NaN	
2	single	sometimes	yes	4.0	school_bus	
3	married	never	no	1.0	NaN	
4	married	sometimes	yes	0.0	school_bus	

	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	< 5	71	71	74
1	5 - 10	69	90	88
2	< 5	87	93	91
3	5 - 10	45	56	42
4	5 - 10	76	78	75

In [7]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30641 entries, 0 to 30640
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            30641 non-null  int64
1   Gender                30641 non-null  object
2   EthnicGroup           28801 non-null  object
3   ParentEduc            28796 non-null  object
4   LunchType             30641 non-null  object
5   TestPrep              28811 non-null  object
6   ParentMaritalStatus   29451 non-null  object
7   PracticeSport         30010 non-null  object
8   IsFirstChild          29737 non-null  object
9   NrSiblings            29069 non-null  float64
10  TransportMeans        27507 non-null  object
11  WklyStudyHours        29686 non-null  object
12  MathScore             30641 non-null  int64
13  ReadingScore          30641 non-null  int64
14  WritingScore          30641 non-null  int64
dtypes: float64(1), int64(4), object(10)
memory usage: 3.5+ MB

```

```
In [10]: df = df.drop(columns=['Unnamed: 0'])
```

```
In [11]: print(df.head(5))
```

	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	\
0	female	NaN	bachelor's degree	standard	none	
1	female	group C	some college	standard	NaN	
2	female	group B	master's degree	standard	none	
3	male	group A	associate's degree	free/reduced	none	
4	male	group C	some college	standard	none	

	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	\
0	married	regularly	yes	3.0	school_bus	
1	married	sometimes	yes	0.0	NaN	
2	single	sometimes	yes	4.0	school_bus	
3	married	never	no	1.0	NaN	
4	married	sometimes	yes	0.0	school_bus	

	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	< 5	71	71	74
1	5 - 10	69	90	88
2	< 5	87	93	91
3	5 - 10	45	56	42
4	5 - 10	76	78	75

```
In [46]: df.isnull().sum()
```

```
Out[46]: Gender          0
         EthnicGroup     0
         ParentEduc      0
         LunchType       0
         TestPrep        0
         ParentMaritalStatus 1190
         PracticeSport   631
         IsFirstChild    904
         NrSiblings     1572
         TransportMeans  3134
         WklyStudyHours   955
         MathScore       0
         ReadingScore    0
         WritingScore    0
         dtype: int64
```

```
In [47]: df['EthnicGroup'] = df['EthnicGroup'].fillna('not specified')
         print(df['EthnicGroup'].value_counts())
```

```
EthnicGroup
group C      9212
group D      7503
group B      5826
group E      4041
group A      2219
no group     1840
Name: count, dtype: int64
```

```
In [62]: df.head(5)
```

```
Out[62]:
```

	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus
0	female	no group	bachelor's degree	standard	none	married
1	female	group C	some college	standard	not specified	married
2	female	group B	master's degree	standard	none	single
3	male	group A	associate's degree	free/reduced	none	married
4	male	group C	some college	standard	none	married

```
In [41]: df['TestPrep'] = df['TestPrep'].fillna('not specified')
         print(df['TestPrep'].value_counts())
```

```
TestPrep
none          18856
completed     9955
not specified  1830
Name: count, dtype: int64
```

```
In [48]: df['ParentMaritalStatus'] =df['ParentMaritalStatus'].fillna('un known')
print(df['ParentMaritalStatus'].value_counts())
```

```
ParentMaritalStatus
married      16844
single       7097
divorced     4919
un known     1190
widowed       591
Name: count, dtype: int64
```

```
In [ ]: df['ParentEduc'] =df['ParentEduc'].fillna('not specified')
print(df['ParentEduc'].value_counts())
```

```
In [51]: df['WklyStudyHours'] =df['WklyStudyHours'].fillna(0)
print(df['WklyStudyHours'].value_counts())
```

```
WklyStudyHours
5 - 10      16246
< 5         8238
> 10        5202
0           955
Name: count, dtype: int64
```

```
In [59]: df['IsFirstChild'] =df['IsFirstChild'].fillna('not applicable')
df['PracticeSport'] =df['PracticeSport'].fillna('NA')
df['NrSiblings'] =df['NrSiblings'].fillna('sibilings not specified')
df['TransportMeans'] =df['TransportMeans'].fillna('Transport not specified')
print(df['IsFirstChild'].value_counts())
```

```
IsFirstChild
yes          19082
no           10655
not applicable    904
Name: count, dtype: int64
```

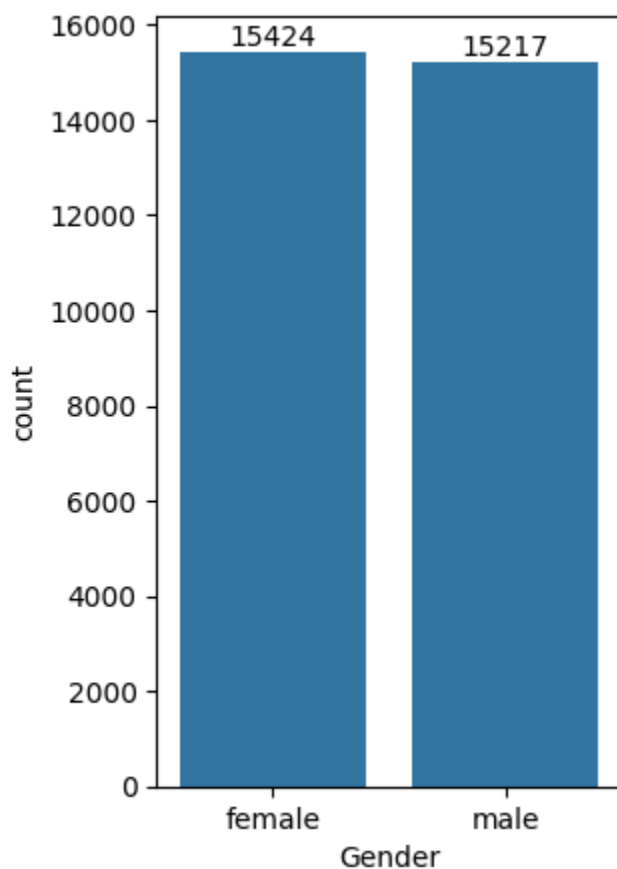
```
In [61]: df.duplicated().sum()
```

```
Out[61]: 0
```

```
In [60]: df.isnull().sum()
```

```
Out[60]: Gender          0
         EthnicGroup      0
         ParentEduc       0
         LunchType        0
         TestPrep         0
         ParentMaritalStatus 0
         PracticeSport    0
         IsFirstChild     0
         NrSiblings       0
         TransportMeans   0
         WklyStudyHours   0
         MathScore        0
         ReadingScore     0
         WritingScore     0
         dtype: int64
```

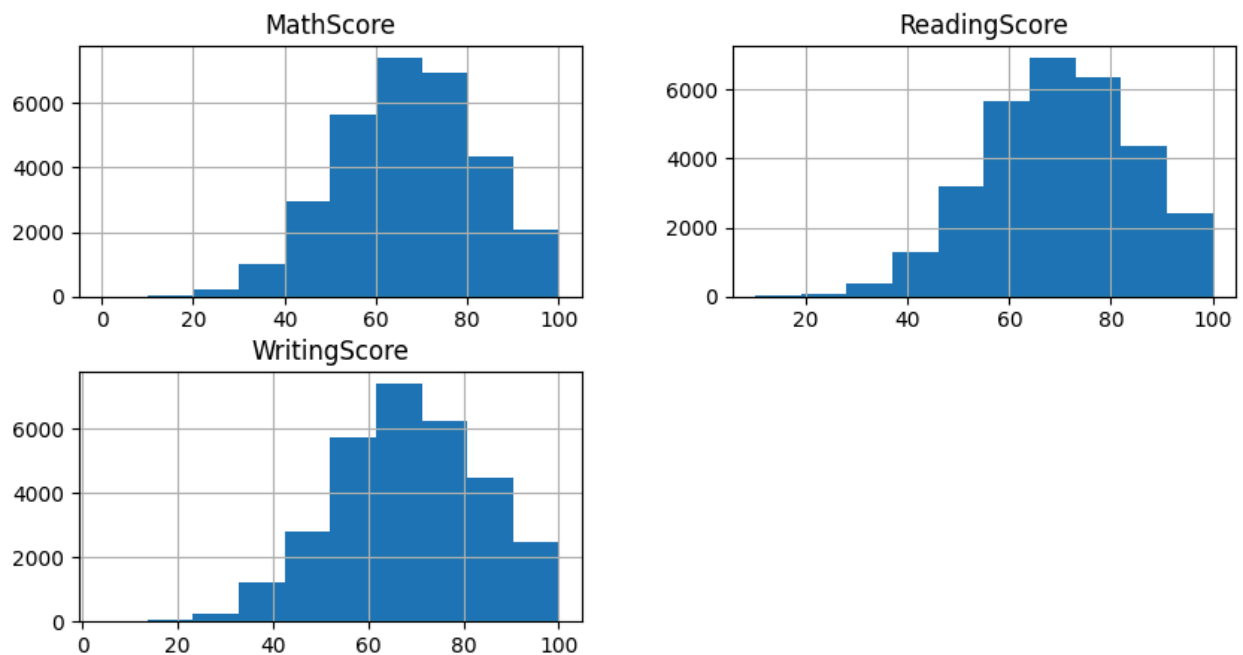
```
In [66]: plt.figure(figsize=(3,5))
         count = sns.countplot(data = df, x="Gender")
         count.bar_label(count.containers[0])
         plt.show()
```



```
In [22]: ab = df.groupby("ParentEduc").agg({"MathScore": "mean", "ReadingScore": "mean", "WritingScore": "mean"})
         print(ab)
```

	MathScore	ReadingScore	WritingScore
ParentEduc			
associate's degree	68.365586	71.124324	70.299099
bachelor's degree	70.466627	73.062020	73.331069
high school	64.435731	67.213997	65.421136
master's degree	72.336134	75.832921	76.356896
some college	66.390472	69.179708	68.501432
some high school	62.584013	65.510785	63.632409

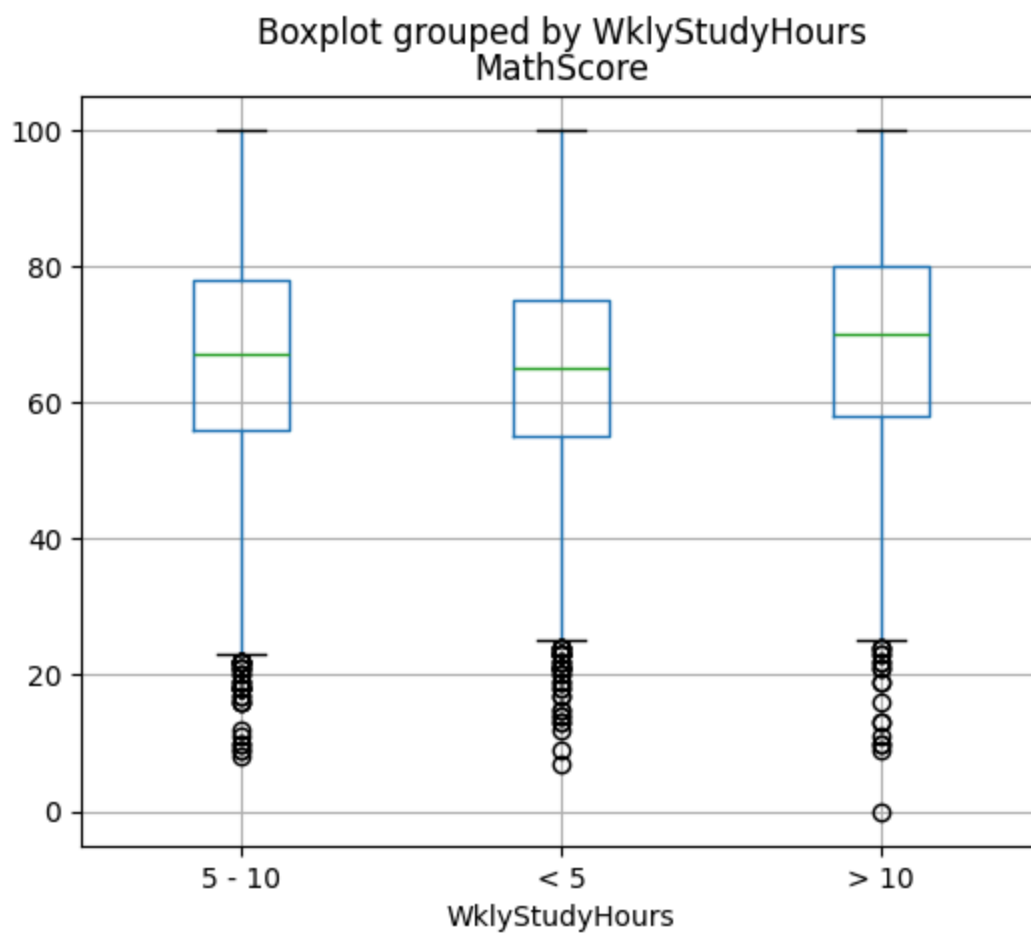
```
In [24]: df[['MathScore', 'ReadingScore', 'WritingScore']].hist(figsize=(10,5))
plt.show()
```



```
In [31]: df.boxplot(column='MathScore', by='Gender', figsize=(6,5))
plt.title("Gender vs Maths score")
plt.show()
```



```
In [28]: df.boxplot(column='MathScore', by='WklyStudyHours', figsize=(6,5))  
plt.show()
```

```
In [64]: df.to_excel("cleaned_dataset student_exam.xlsx", index=False)
```

```
In [ ]:
```